Problem

A certain store contains three balanced three-phase loads. The three loads are:

Load 1: 16 kVA at 0.85 pf lagging

Load 2: 12 kVA at 0.6 pf lagging

Load 3: 8 kW at unity pf

The line voltage at the load is 208 V rms at 60 Hz, and the line impedance is 0.4 + *j*0.8 Ω. Determine the line current and the complex power delivered to the loads.

Step-by-step solution

Step 1 of 7

Substitute $31.79°$ for θ_1 and $16\,\text{kVA}$ for S_1 in equation (1).

$$\begin{aligned}\overline{S_1} &= (16000)\cos(31.79°) + j(16000)\sin(31.79°) \\ &= (16000)(0.849) + j(16000)(0.526) \\ &= 13.58\,\text{kVA} + j8.41\,\text{kVA} \\ &= (13.6 + j8.43)\,\text{kVA}\end{aligned}$$

Therefore, the complex power delivered to the load 1 is $\boxed{(13.6 + j8.43)\,\text{kVA}}$.

Step 2 of 7

Write the formula for complex power to load 2.

$$\overline{S_2} = S_2\cos\theta_2 + jS_2\sin\theta_2 \quad \text{......} \ (2)$$

Where,

The power factor, $\cos\theta_2$

For load 2, the pf is 0.6. Therefore,

$$\begin{aligned}\cos\theta_2 &= 0.6 \\ \theta_2 &= \cos^{-1}(0.6) \\ \theta_2 &= 53.13°\end{aligned}$$

Step 3 of 7

Substitute $53.13°$ for θ_2 and $12\,\text{kVA}$ for S_2 in equation (2).

$$\begin{aligned}\overline{S_2} &= (12000)\cos(53.13°) + j(12000)\sin(53.13°) \\ &= (12000)(0.6) + j(12000)(0.799) \\ &= 7.2\,\text{kVA} + j9.6\,\text{kVA} \\ &= (7.2 + j9.6)\,\text{kVA}\end{aligned}$$

Therefore, the complex power delivered to the load 2 is $\boxed{(7.2 + j9.6)\,\text{kVA}}$.

Step 4 of 7

Write the formula for complex power to load 3.

$\overline{S_3} = S_3 \cos\theta_3 + jS_3 \sin\theta_3$ ...... (3)

Where,

The power factor, $\cos\theta_3$

For load 3, the pf is 1. Therefore,

$$\begin{aligned}
\cos\theta_3 &= 1 \\
\theta_3 &= \cos^{-1}(1) \\
\theta_3 &= 0°
\end{aligned}$$

Step 5 of 7

Substitute $0°$ for θ_3 and $8\ \text{kVA}$ for S_3 in equation (3).

$$\begin{aligned}
\overline{S_3} &= (8000)\cos(0°) + j(8000)\sin(0°) \\
&= (8000)(1) + j(8000)(0) \\
&= 8\ \text{kVA} + j0\ \text{kVA} \\
&= (8 + j0)\ \text{kVA}
\end{aligned}$$

Therefore, the complex power delivered to the load 2 is $\boxed{(8 + j0)\ \text{kVA}}$.

Step 6 of 7

Determine the total complex power of the loads by adding the three complex powers.

$$\begin{aligned}
\mathbf{S}_P &= \overline{S_1} + \overline{S_2} + \overline{S_2} \\
&= \left[(13.6 + j8.43) + (7.2 + j9.6) + (8 + j0)\right]\text{kVA} \\
&= 28.8 + j18.03\ \text{kVA}
\end{aligned}$$

The source voltage, $\mathbf{V}_{ab} = 208\ \text{V rms}$

Calculate the value of the $\mathbf{V}_{an}$.

$$\begin{aligned}
\mathbf{V}_{an} &= \frac{\mathbf{V}_{ab}\angle -30°}{\sqrt{3}} \\
&= \frac{208\angle -30°}{\sqrt{3}} \\
&= 120.08\angle -30°
\end{aligned}$$

It is known that the complex power is,

$\mathbf{S}_P = \mathbf{V}_P \mathbf{I}_P^*$ ...... (4)

Where,

The current $\mathbf{I}_P = \mathbf{I}_L$

Rearrange the equation (4) for $\mathbf{I}_P^*$.

$$\begin{aligned}
\mathbf{I}_P^* &= \frac{\mathbf{S}_P}{\mathbf{V}_P} \\
&= \frac{28800 + j18030}{(3)(120.08)} \\
&= 79.95 + j50.05\ \text{A}
\end{aligned}$$

$$\begin{aligned}
\mathbf{I}_L &= 79.95 - j50.05\ \text{A} \\
&= 94.32\angle 32.05°\ \text{A}
\end{aligned}$$

Step 7 of 7

Now, assume the positive phase rotation. But the voltage $\mathbf{V}_{ab} = 208\angle 0°$ then $\mathbf{V}_{an} = 120.08\angle -30°\ \mathrm{V}$. Then which yields,

$$\begin{aligned}\mathbf{I}_a &= 94.32\angle -30° - 32.05°\ \mathrm{A}\\ &= 94.32\angle -62.05°\ \mathrm{A}\\ \mathbf{I}_b &= 94.32\angle -120° - 30° - 32.05°\ \mathrm{A}\\ &= 94.32\angle 177.95°\ \mathrm{A}\\ \mathbf{I}_c &= 94.32\angle 120° - 30° - 32.05°\ \mathrm{A}\\ &= 94.32\angle 57.95°\ \mathrm{A}\end{aligned}$$

Therefore, the line currents $\mathbf{I}_a, \mathbf{I}_b, \mathbf{I}_c$ are $\boxed{94.32\angle -62.05°\ \mathrm{A}}$, $\boxed{94.32\angle 177.95°\ \mathrm{A}}$ and $\boxed{94.32\angle 57.95°\ \mathrm{A}}$ respectively.